

SIMATS ENGINEERING
CO2_ASSESSMENT TOOL3_Application Based Questions

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA1522

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COURSE OUTCOME COVERED

CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct

I Brijesh.T/192521391 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: 

Assessment Tasks

Application Based Questions

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.
2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.
3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge

servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

4. A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.
5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

Application Based Questions Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Relevance to Application	25%	Response directly addresses the given use case with strong relevance	Mostly relevant response	Partially relevant	Weak relevance
Use of Cloud Concepts	25%	Applies appropriate concepts accurately	Mostly accurate application	Basic application only	Incorrect concept usage
Practical Insight	20%	Shows strong real-world understanding	Shows reasonable practical understanding	Limited practical insight	No practical insight
Completeness	15%	Response covers all required aspects	Covers most aspects	Covers only basic aspects	Incomplete response
Clarity	15%	Clear and concise answer	Generally clear	Some confusion in expression	Poor clarity

1. Cloud Auto-Scaling and Load Balancing

A company with seasonal sales often experiences sudden increases in website traffic. Cloud auto-scaling automatically adds more servers or virtual machines when traffic increases and removes them when demand decreases. This ensures the application always has enough resources without wasting money during normal periods.

Load balancing distributes incoming user requests across multiple servers. Instead of one server handling all requests, traffic is shared equally, preventing overload and improving performance.

Benefits

Reduces downtime by preventing server overload.

Improves user experience with faster response times.

Saves cost by using resources only when needed.

Ensures high availability during peak traffic.

Role of Monitoring Tools

Monitoring tools continuously track:

CPU and memory usage

Network traffic

Response time

Server health

If resource usage crosses a threshold, the monitoring system triggers auto-scaling to add more instances automatically.

Real-World Example

During the Amazon Prime Day sale, millions of customers visit the website simultaneously. Auto-scaling launches additional servers, while load balancers distribute customer requests evenly, ensuring smooth shopping without website crashes.

2. Cloud Storage for Backup Data

Cloud storage provides secure and reliable storage for large amounts of backup data. Cloud providers automatically store multiple copies of data across different servers and locations.

Durability and Data Redundancy

Data is replicated across multiple data centers.

If one server fails, another copy is immediately available.

This provides very high durability and availability.

Security Measures

Encryption

Encrypts data before storage and during transmission.

Prevents unauthorized users from reading backup files.

Access Control

Uses Identity and Access Management (IAM).

Only authorized users can access or restore backups.

Supports Multi-Factor Authentication (MFA).

Cost Advantages

Compared to traditional storage:

No need to purchase expensive storage hardware.

No maintenance costs.

Pay only for the storage used.

Easy to increase or decrease storage capacity.

Disaster Recovery Example

A company stores daily backups in the cloud. If a flood destroys its local servers, the backup can be restored from cloud storage, allowing business operations to continue with minimal downtime.

3. Content Delivery Networks (CDNs)

A Content Delivery Network (CDN) stores copies of website content at multiple locations worldwide, allowing users to access data from the nearest server.

Benefits

Faster loading times

Lower network latency

Reduced server workload

Better user experience

Role of Edge Servers

Edge servers are located close to users. They:

Deliver cached content quickly.

Reduce the distance data travels.

Improve performance in different countries.

Global Availability

Cloud providers deploy data centers across multiple regions. If one region experiences an outage, traffic is redirected to another available region, ensuring uninterrupted service.

Real-Time Example

A global video streaming platform like Netflix uses CDNs to deliver movies from nearby edge servers. Users receive smooth video playback with minimal buffering regardless of their location.

4. Cloud Security and Regulatory Compliance

Financial institutions must protect customer information while meeting government regulations.

Regulatory Compliance

Cloud providers support standards such as:

PCI DSS

ISO 27001

SOC 2

GDPR (where applicable)

These certifications help organizations meet legal and security requirements.

Role of Logging and Monitoring

Logging records:

User logins

Data access

System changes

Security events

Monitoring continuously checks for suspicious activities and generates alerts when unusual behavior is detected.

Role of Encryption

Encrypts customer information during storage.

Encrypts transactions while data travels across the internet.

Protects confidential financial records.

Data Sovereignty

Organizations can store data in specific geographic regions to comply with national laws that require customer data to remain within a particular country.

Example

A bank processes online fund transfers. Every transaction is encrypted, logged, and monitored. Audit logs help investigators verify all financial activities while customer data remains stored within approved regional data centers.

5. Containers and Kubernetes

Containers package an application with all its dependencies so it can run consistently on any system.

Containers vs Virtual Machines

Containers

Virtual Machines

Lightweight

Heavyweight

Share host operating system

Each VM has its own operating system

Start in seconds

Take longer to start

Lower resource usage

Higher CPU and memory usage

Higher application density

Lower application density

Kubernetes Orchestration

Kubernetes automates:

Container deployment

Scaling

Load balancing

Self-healing

Rolling updates

If a container fails, Kubernetes automatically replaces it without affecting users.

Portability

Containers run consistently across:

Private cloud

Public cloud

Hybrid cloud

Local development environments

This makes application migration between cloud providers much easier.

Microservices Example

An online shopping application can be divided into independent services:

User Service

Product Catalog Service

Shopping Cart Service

Payment Service

Order Service

Each service runs in its own container. Kubernetes manages all containers, automatically scaling busy services like the Product Catalog during high customer demand while keeping the application highly available.